

Week 1: ANSWERS

1 (a) Division by 0: e.g. if ℓ is parallel to the x -axis, then $p = 0$.

(b) $y = -\frac{q}{p}x + q$, the slope is $-\frac{q}{p}$.

(c) 11.

(d) $p_k = \frac{k}{10}$, $q_k = \frac{10-k}{10} = 1 - \frac{k}{10}$.

(e) $y = \frac{k-10}{k}x + \frac{10-k}{10}$.

2 (a) $\frac{x}{p} + \frac{y}{1-p} = 1$.

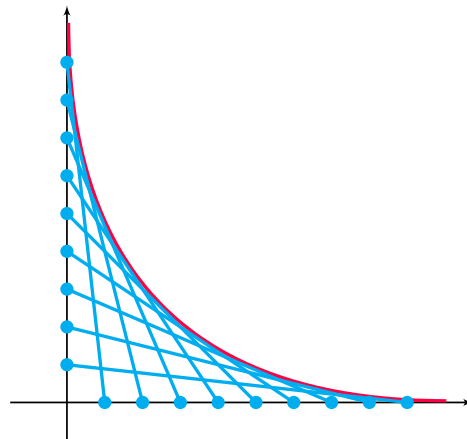
(b) $\begin{cases} E: & u + v = 1, \\ \ell: & \frac{u^2}{p} + \frac{v^2}{1-p} = 1. \end{cases}$

(c) $v = 1 - u$,

Quadratic equation: $\frac{1}{p(1-p)}u^2 - \frac{2}{1-p}u + \frac{p}{1-p} = 0$, simplified by multiplying with $p(1-p)$: $u^2 - 2pu + p^2 = 0$.

Solution: $u = p$, hence $v = 1 - p$.

(d) $x = p^2$ and $y = (1-p)^2$.



3 If $\frac{R}{r} = \frac{m}{n}$ with m and n positive integers, then $\theta_{\max} = \frac{2n\pi}{\gcd(m,n)}$, where

$\gcd(m, n)$ is the greatest common divisor of m and n .

4 (a) $x^4 - 2x^2 + 1 + 2x^2y^2 + 2y^2 + y^4 = b^4$.

(b) First dots: -2 , second dots: $1 - b^4$, so the equation becomes

$$(x^2 + y^2)^2 - 2(x^2 - y^2) + 1 - b^4 = 0. \quad (1)$$

(c) $y = \pm\sqrt{b^2 - 1}$

- if $b > 1$ then there are two intersection points.
- if $b = 1$ then there is one intersection point.
- if $b < 1$ then there are no intersection points.

(d) $x = \pm\sqrt{1 \pm b^2}$

- if $b > 1$ then there are two intersection points ($x = \pm\sqrt{1 + b^2}$).
- if $b = 1$ then there are three intersection points ($x = 0$ and $x = \pm\sqrt{2}$).
- if $b < 1$ then there are four intersection points ($x = \pm\sqrt{1 + b^2}$ and $x = \pm\sqrt{1 - b^2}$).

(e) $x^2 + y^2 = r^2$ and $x^2 + y^2 = r^2 \cos(2\theta)$. From (1) follows:

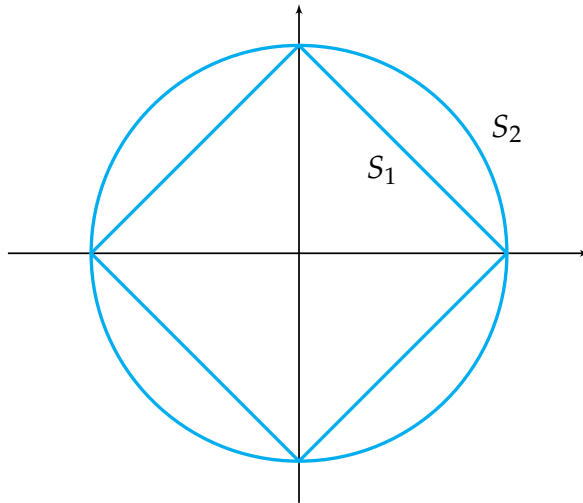
$$r^4 - 2r^2 \cos(2\theta) + 1 - b^4 = 0.$$

(f) $w^2 - 2w \cos(2\theta) + 1 - b^4 = 0$.

(g) $r(\theta) = \sqrt{w} = \sqrt{\cos(2\theta) \pm \sqrt{b^4 - \sin^2(2\theta)}}$.

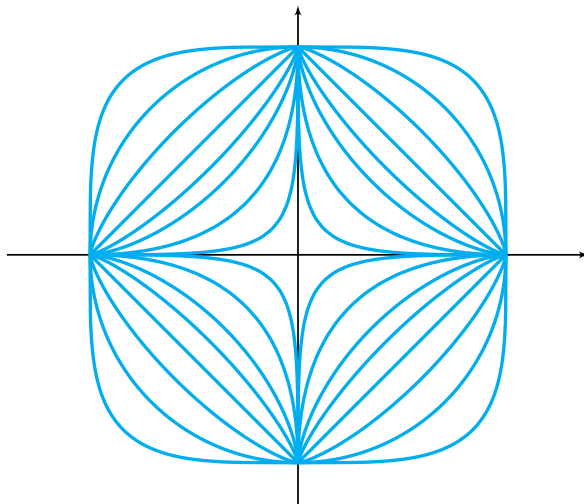
6 (a) $S_{1/2}$ is the envelope of the lines in string art (see exercise 2).

(b) S_1 is a square, S_2 is a circle:



(c) $\alpha = \frac{2}{\mu}$.

(d) With $\mu = 0.5, 1, 1.5, 2, 2.5, 3, 4, 6$:



(e) S_0 is a cross, consisting of two line segments, from $(-1,0)$ to $(1,0)$ and from $(0,-1)$ to $(0,1)$ respectively.

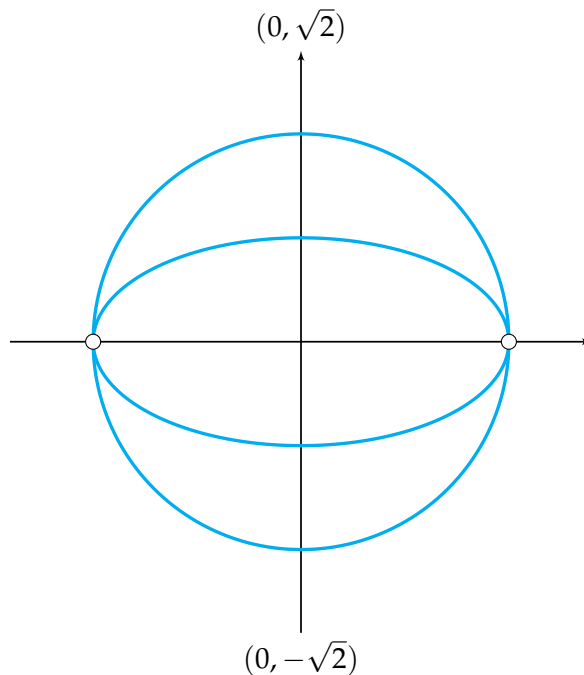
(f) S_∞ is a 2×2 -square with vertical and horizontal edges, centered at $(0,0)$.

7 Contours: $x^2 + y^2 = C$.

- If $C > 0$ then the contour is a circle with radius $\sqrt{C}$.

- If $C = 0$ then the contour consists of a single point (the origin).²
- If $C < 0$ then the contour is the empty set.

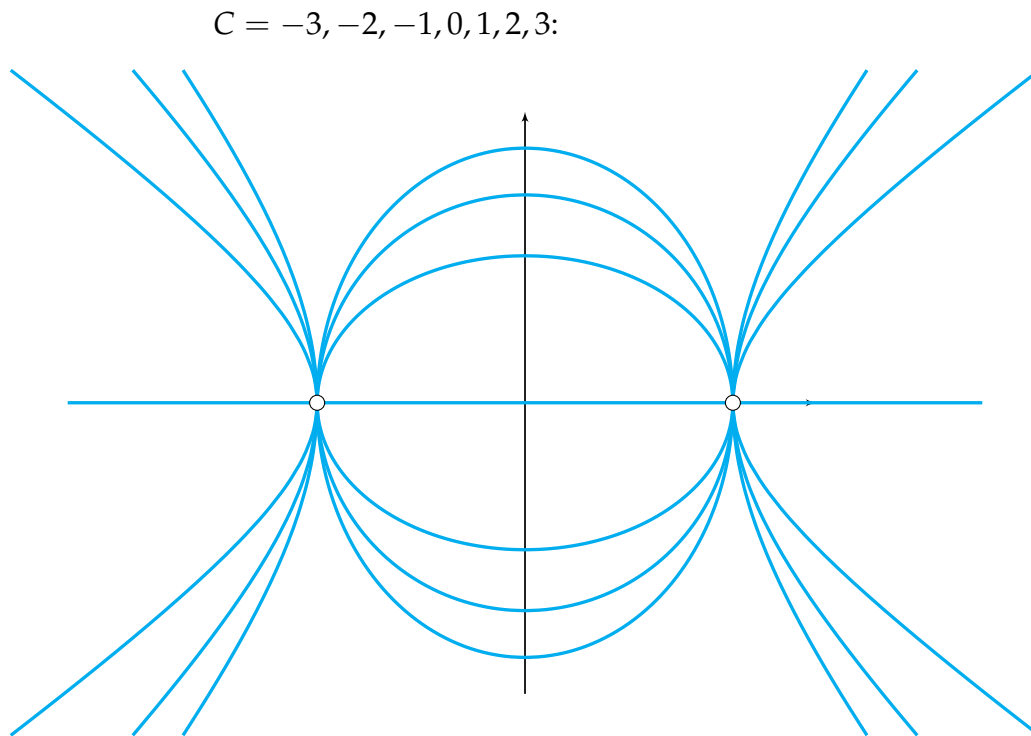
- 8 (a) f is not defined when $x = 0$, so points of the form $(\pm 1, y)$ for $y \in \mathbb{R}$ are always excluded. If $C = 2$, we end up with $2x^2 + 2y^2 = 2$ or $x^2 + y^2 = 1$, which is the equation for the unit circle, so the contour is the circle minus the points $(1, 0)$ and $(-1, 0)$. If $C = 1/2$ we get $x^2 + 4y^2 = 1$ instead, so the contour is an ellipse with vertical semiaxis $1/2$ and horizontal semiaxis 1 with the points $(1, 0)$ and $(-1, 0)$ excluded.



- (b) If $C > 0$, as above the contours are ellipses with vertical semiaxis $\sqrt{C}/2$ and horizontal semiaxis 1 minus the the points $(1, 0)$ and $(-1, 0)$. If $C = 0$, the contour is the horizontal axis minus the the points $(1, 0)$ and $(-1, 0)$. If $C < 0$, $2y^2 + Cx^2 = C$ or $x^2 = 1 - 2y^2/C$ is satisfied by two hyperbolas. These don't contain any points of the vertical axis, so the contours are the complete hyperbolas.

In conclusion, the contours are not empty for all $C \in \mathbb{R}$. Plotting with

²You could argue that this is a circle with radius 0 .

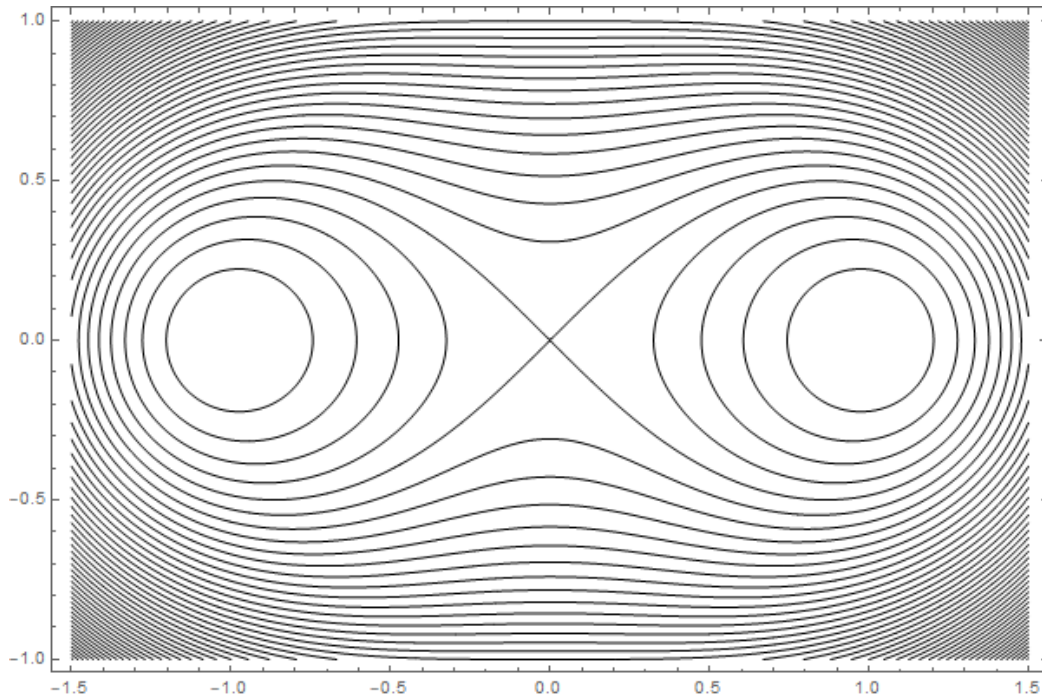


9 (a) A crater.

(b) Somewhere in between -0.8 m and -1.0 m^3 .

10 $f(x, y) = (x^2 + y^2)^2 - 2(x^2 - y^2) + 1.$

³NASA uses SI units ever since the [failure in 1999](#) of the Mars Climate Orbiter.



- 13 For any point z , first we translate so that $(1, 1) = 1 + i$ goes to the origin, then rotate, and finally translate back:

$$e^{i\frac{\pi}{4}}(z - (1 + i)) + (1 + i) = e^{i\frac{\pi}{4}}(z - \sqrt{2}e^{i\frac{\pi}{4}}) + \sqrt{2}e^{i\frac{\pi}{4}} = e^{i\frac{\pi}{4}}z + \sqrt{2}(e^{i\frac{\pi}{4}} - e^{i\frac{\pi}{2}}).$$

Remembering $\cos \pi/4 = \sin \pi/4 = \frac{1}{2}\sqrt{2}$ we can write this as:

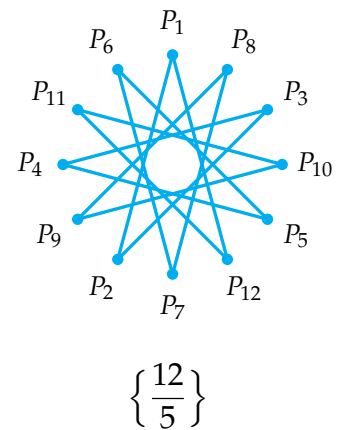
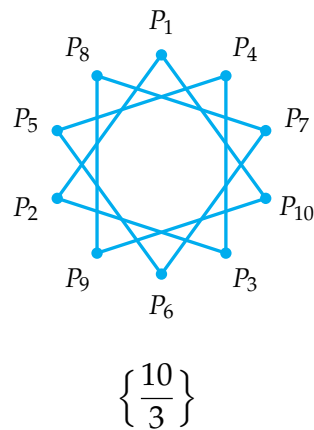
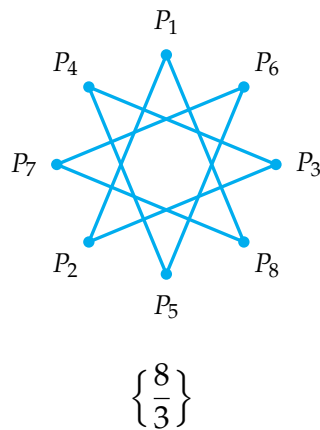
$$\begin{aligned} e^{i\frac{\pi}{4}}z + \sqrt{2}(e^{i\frac{\pi}{4}} - e^{i\frac{\pi}{2}}) &= \left(\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}\right)z + \sqrt{2}\left(\frac{\sqrt{2}}{2} + \left(\frac{\sqrt{2}}{2} - 1\right)i\right) \\ &= \left(\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}\right)z + \left(1 + \left(1 - \frac{\sqrt{2}}{2}\right)i\right). \end{aligned}$$

- 14 The circle with center $(0, -\frac{1}{2})$ and radius $\frac{1}{2}$. One can check (with some work) that for any $x \in \mathbb{R}$ we have

$$\left(\frac{x}{x^2 + 1}\right)^2 + \left(\frac{-1}{x^2 + 1} + \frac{1}{2}\right)^2 = \dots = \frac{1}{4}.$$

- 15
- Polygon (c) is a 10-gon, all other polygons are 5-gons.
 - They are all equilateral.
 - Polygons (a) and (b) are equiangular.

16 (a)



- (b) The angle at a vertex of $\{5\}$ is 108° .
 The angle at a vertex of $\left\{ \frac{10}{3} \right\}$ is 72° .